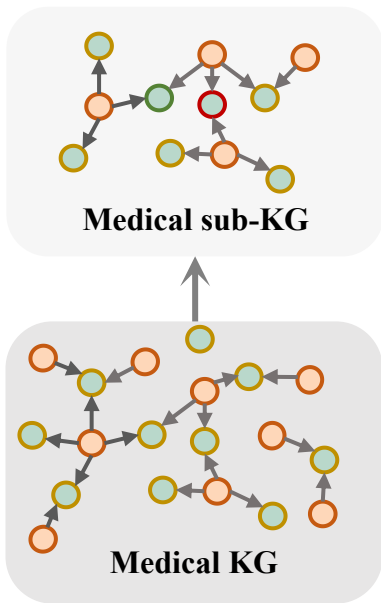
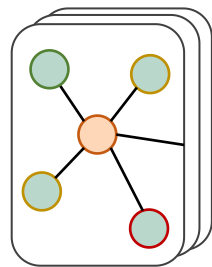


Medical Subgraph Extraction

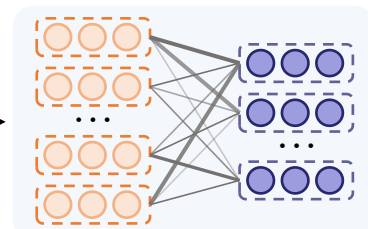
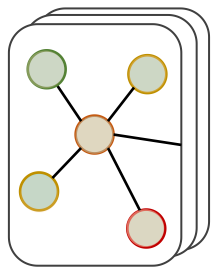


Question: I feel nauseous and vomiting, accompanied by fever, diarrhea, and loose stools. What's going on?



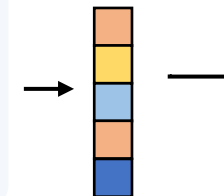
Local Semantic Information Aggregation

GCN



Semantic Similarity Calculation

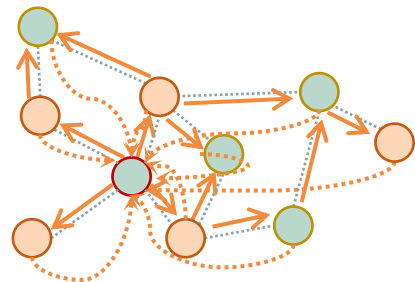
Feedback from KG



Semantic Relevance Scores

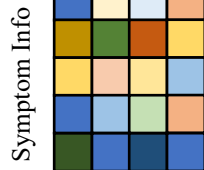
+

KG
Feedback

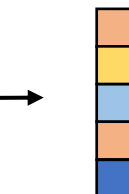


Global Topological Link Walk

Disease Prediction



Reachability Matrix



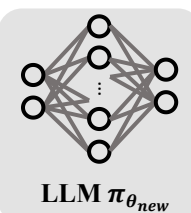
Link Criticality Scores

LLM Reinforcement Learning



PPO

$$\min(\frac{\pi_{\theta_{new}}}{\pi_{\theta_{old}}} \hat{A}, \text{clip}(\frac{\pi_{\theta_{new}}}{\pi_{\theta_{old}}}, 1 - \epsilon, 1 + \epsilon) \hat{A})$$



Response: The patient's nausea, vomiting, and loose stools suggest a gastrointestinal issue, likely gastroenteritis, given the fever and diarrhea.